

From Protest to Policy:
How Child Safety Concerns Catalyzed a Sustainable Mobility Movement in the
Netherlands

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Intro

Road crashes are the leading killer of children and youth worldwide and are the 12th leading cause of death when all ages are considered, according to the World Health Organization (2023). This is a major health crisis that goes relatively under acknowledged because the public is desensitized and dismissive, viewing traffic fatalities as inexorable or due to individual circumstances. This is even noticeable down to the way we euphemize traffic crashes as merely traffic *accidents*. The car fatality has shifted from a “moral wrong to a necessary evil” (Vardi 2014, p.346). Are these deaths not just the risks that come along with the car-centric world we live in? Not at all. The costs of car-centric urban planning extend far beyond collision statistics: they include chronic air pollution, soaring emissions, inaccessible urban spaces, and the erosion of public life. These consequences are what make car-centric urban spaces fundamentally unsustainable. The need to move away from car dependency is not only about safety, but also about building healthier, more equitable, and more sustainable cities. Even without switching modes of transport, we can make vehicles safer through altering the built environment. One small country in western Europe had figured this out after the people were fed up with car-related deaths. The people, government, and engineers came together to create a fundamentally safer transport network that shifted the focus away from cars.

This research paper explores how public outrage over child road deaths in the 1970s Netherlands catalyzed a systemic shift away from car-centric urban planning and toward sustainable, people-first transportation design. This paper begins by evaluating the

sustainability of cycling using the triple bottom line framework. It then outlines the problems associated with unsustainable, car-centric hypermobility and explores the historical conditions that led to the rise of the automobile in the Netherlands. The paper next examines the key events that triggered a shift in urban planning and illustrates how that transformation took shape. Finally, it analyzes the outcomes of this shift, showing what safe and livable streets can look like when human life is prioritized over car dominance.

The Sustainability of Cycling

While the normalization of traffic deaths is perhaps the most tragic consequence of car dependency, it is only one part of a much broader pattern of harm. To fully understand why car-centric planning is unsustainable, we must examine its wider social, environmental, and spatial costs. Before examining the Dutch case in detail, it is essential to define what sustainable transportation is and why it matters and why the bike falls into this category. The Netherlands is only discussed in this paper precisely because it offers a leading example of sustainable mobility in practice. Sustainable transportation is the epicenter of environmental responsibility, social equity, and economic vitality. The triple

bottom line framework of social, environmental, and economic as illustrated by Tumlin in Figure 1 will be used in this analysis.

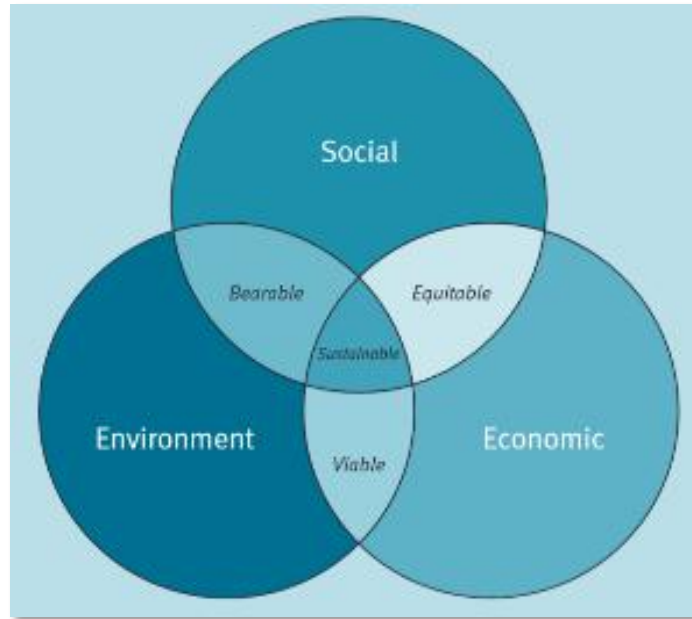


Figure 1. The triple bottom line and its intersections. (Tumlin, 2012)

Environmental

The need for action in the form of a sustainable transportation forward policy in response to the climate crisis has never been more pressing than right now. The scientific community is in unison urging our policy makers about the dangers of anthropogenic carbon dioxide emissions and the “irrevocable sea level rise,” “increased fire frequency,” and “long-term rainfall decreases” that will follow (Solomon et al. 2009). The emissions of hundreds of millions of cars on the road each day no doubt contribute immensely to this problem. What was once a grand solution to the average person and a boon to the economy has become a driving force behind one of the most urgent crises of our time. Climate

change. The bike is a tangible solution to our crisis. According to Brand et al. (2021), an individual who shifts from car to bicycle travel reduces their life cycle CO₂ emissions by an average of 3.2 kg per day, highlighting the significant climate benefits of even modest changes in travel behavior. Environmentally, the bike only requires the initial raw materials of metal and rubber for production and does not require any emission-emitting energy source. It contributes no harmful emissions to the air and emits virtually no noise pollution.

Social

Bicycling is inherently socially sustainable because it provides an affordable and accessible means of transportation for a broad range of people while having a much lower barrier to entry than a car does. Even at its inception, it has been equitable. “Cycling was the first popular athletic activity open to women on a large scale,” which helped progress the feminist movement of the late 1800s (Schiller and Kenworthy, 2018, p. 87). Early Dutch cycling clubs used rhetoric that posited cycling was for the Dutch of all ages, genders, and income levels. These clubs used the bicycle as a national icon to symbolize unity, health, and modernity (Ebert 2004). Cycling’s affordability allows people from lower-income backgrounds to travel independently and reliably without the burden of expensive vehicle-related expenses. Cycling also does not require the same hurdles as taking driving lessons and the multiple driving exams required to own a driver’s license. Those who are not of the same ability as the average person can look to electronic bikes as a form of participating in sustainable mobility (Van Der Craats et al. 2025). In contrast, car-centric planning is

socially unsustainable because it leads to a “lack of community feeling” and a “poor sense of belonging” to their neighborhood (Schiller and Kenworthy 2018, p.26).

Economic

From an economic perspective, bicycling offers significant advantages that contribute to the overall sustainability of transportation systems. The low cost of purchasing and maintaining a bicycle makes it an accessible option for individuals across income levels, reducing financial barriers to mobility. Unlike car ownership, which involves expenses such as fuel, insurance, parking fees, and costly repairs, bicycles require minimal upkeep and no fuel, translating to substantial savings for users. Beyond individual benefits, widespread cycling reduces the strain on public infrastructure by lessening wear and tear on roads and lowering demand for expensive parking facilities. This decrease in infrastructure costs benefits municipalities and taxpayers alike. Furthermore, by supporting local economies, through increased accessibility to businesses and encouraging active lifestyles that reduce healthcare costs, bicycling fosters economic vitality at both the community, commercial, and societal levels. A meta-analysis found that replacing car parking with cycling infrastructure often benefits local economies. Of the bicycle facilities studied, 57% had positive economic effects, while 29% showed no significant impact (Volker and Handy 2021). In these ways, biking promotes economic sustainability by enabling affordable, efficient, and community-centered transportation.

Biking stands at the heart of sustainable transportation because it simultaneously fulfills each of the three pillars of ecological, social, and economic goals. As Tumlin exclaims

in his book *Sustainable Transportation Planning*, the bicycle is “the most energy-efficient form of transportation ever invented,” while also being “one of the most space efficient” modes of transport (Tumlin, 2012, p.73). The bicycle is not merely a symbol of sustainable transportation; it is its most effective embodiment. Its minimal environmental footprint, inherent social inclusivity, and affordability make it uniquely positioned to address the urgent challenges posed by car dependency. As the next section will show, the consequences of neglecting active transportation like cycling reveal a deeper public health crisis hiding in plain sight.

History & Context

Rise of the Bicycle

A socio-technical analysis of Dutch transportation history reveals how cultural values, political institutions, and grassroots advocacy co-evolved with technology to shape the country's deep-rooted cycling culture and later confrontations with car-centric planning. Once the bike was introduced to the Netherlands in 1870, the upper-class Dutch were quick to pick up the new mode of transport. It was not until the bourgeoisie drove high demand for bicycles in the 1890s that economies of scale would allow for prices to plummet (Ebert 2004). The bicycle was being sold as a new technology that would allow the Dutch to be “in control again” and be their “own master” when riding (Ebert, 2004, p. 352). Since the high-class folk had picked up the bicycle first, it was seen to all as a status symbol, enticing the Dutch to join the society of cyclists. As the middle and lower classes were introduced to this invention, the Dutch began to see the bicycle more as a means of

transport rather than a leisure activity or hobby. Cyclists formed organizations to band together against the rural Dutch who saw the bicycle as a nuisance and harassed cyclists whenever they could; some even throwing “heavy chunk[s] of coal” at passing by cyclists (Dutch Ministry of Transport, Public Works and Water Management 1999). These clubs later formed into an official organization known as the General Dutch Cyclist Union (abbreviated to ANWB), which would prove indispensable in providing cyclists with a voice within the government. Their formation helped legitimize cycling as a practical and respectable mode of transportation, rather than a trend or novelty.

The expansive access to government, Dekker (2021) argues, was due to a unique facet of Dutch culture known as the Polder Model. The polder model is the Dutch tradition of consensus-based decision-making between government, businesses, and unions, formalized during the 1982 Wassenaar Accord in response to the economic crisis. Rooted in the Netherlands' historical need for cooperative water management, the model emphasizes collaboration, shared responsibility, and pragmatic compromise to address national challenges (Schreuder 2001). Although the term “polder model” wasn’t coined until 1982, its principles were evident much earlier, as far back as the Middle Ages (De Vries 2014). The influence of this model can be seen in the development of early cycling policy, particularly through the ANWB’s active lobbying. This spirit of inclusion meant that policymaking was not solely the domain of elected officials and technocrats, but also actively engaged civil society organizations, advocacy groups, and even individual citizens. Input from the public was regarded not as disruptive, but as essential to informed decision-making and a healthy democratic process.

It was also the connections that early cycling groups had to policymakers that granted them such a large voice. As Dekker (2021) points out, “their leaders ... had connections with (local) political elites, which helped members secure subsidies and permits for building local cycle paths” (p. 68). This early example of collaborative governance between citizens and the state exemplifies how Dutch infrastructural development was never solely a top-down affair. Instead, it reflected a dynamic negotiation between social actors, cultural attitudes, and technological change. The ANWB’s success in influencing early cycling policy demonstrates how civil society groups could effectively shape transportation priorities, long before the automobile entered mainstream use.

Rise of The Auto

Despite these strong cycling roots, the 1920s and 1930s saw the Netherlands increasingly orienting its planning and infrastructure toward cars. In 1923, even though bicycles vastly outnumbered cars 30 to 1, national policy began prioritizing automotive infrastructure, funneling millions of guilders into highways and bridges (Dutch Ministry of Transport, Public Works and Water Management 1999). Just as the bicycle had once been promoted, the car was now being sold to the public using the very same promises: freedom, status, and a comfortable mode of transportation. This was due to the new presence of aggressive car-forward lobbyists within the government. Corporations like Shell and the International Road Federation were creating lobbyist groups to further push the car market forward (Van Der Craats & Van Lierop, n.d.). It was big oil corporations versus the small cyclists’ unions, which often consisted of middle-class Dutch. These cycling clubs had far

fewer resources than the large oil companies, and pleas to stop car infrastructure were ignored. These investments paved the way for what was seen as the inevitable shift toward car-centric mobility, mirroring developments occurring in the United States. The Netherlands, no doubt, saw the automobile as the future and wanted to be ready for a time when everyone was to own one. This resulted in a very large majority of cyclists being made to reluctantly give up “their space to a minority of car drivers” (Dekker 2022). This misguided urban planning approach ignored the possibility that many citizens might continue to prefer cycling in the future, both for its practicality and health benefits and instead sidelined those who chose not to adopt the car.

The post-war Netherlands of the 1950s and 1960s saw a substantial increase in personal motor vehicles and showed no signs of slowing down. After the war, the Netherlands needed time to rebuild the structures that had been leveled, and the United States was providing them with funds to do so. Along with the US aid came the same planning ideals and ideologies (Van Der Craats and Van Lierop, n.d.). The United States had already been quick to drop the bicycle itself (Dutch Ministry of Transport, Public Works and Water Management, 1999). They no doubt were suggesting the Netherlands follow suit. According to the Dutch Ministry of Transport, Public Works and Water Management, some of the main reasons for the large increase in personal auto use between 1950 to 1975 were: an increase in disposable income by a factor of 3, an increase in urban sprawl, and an increase in labor specialization (leading to longer commutes) (1999). The boost in Dutch prosperity, combined with the personal automobile becoming cheaper and more plentiful, led to cars being affordable for the first time ever. The automobile became a sign of success, personal freedom, and modernity. With this rise in prosperity came a rise in population and

housing. Because the planning philosophy at the time was centered around the car, these new residential complexes were being built further away from the city centers. This all made owning a personal automobile a very tempting endeavor. The car had been on Dutch roads for decades now, but only in the 1950s did it see a large increase in users. Between 1955 and 1975, car commuting surged in the Netherlands, leading to a steep decline in public transit use. During this time, the portion of travel done by bus, tram, or train fell from over half of all commuter trips to less than one-fifth (Verlaan, 2021).

The Hidden Health Crisis

Acute Damage

The Dutch did not need decades of research to understand how hazardous cars were. They could simply talk to a neighbor who had lost a child to a car crash. Movements to restrict the car started at the inception of the automobile and came from avid cyclists. As plans to increase motorways unfolded, more citizens recognized that cyclists and pedestrians were getting the short end of the stick. Looking at historical data on road deaths in the Netherlands since 1950 (Figure 2), we see a sharp trend upwards as the personal motor vehicle starts to become widely adopted. Starting at 1000 deaths per year in 1950, road deaths increased to over 3000 per year in the late 1960s and early 1970s. This new, devastatingly horrific cause of death outraged the Dutch public. Many began to realize that the violence of car-centric streets was not just an unfortunate side effect of modern life; rather, it was a direct result of how cities had been planned. Dutch cities had

given too much space over to cars, crowding out pedestrians and cyclists and transforming once-vibrant public spaces into congested, dangerous roadways.

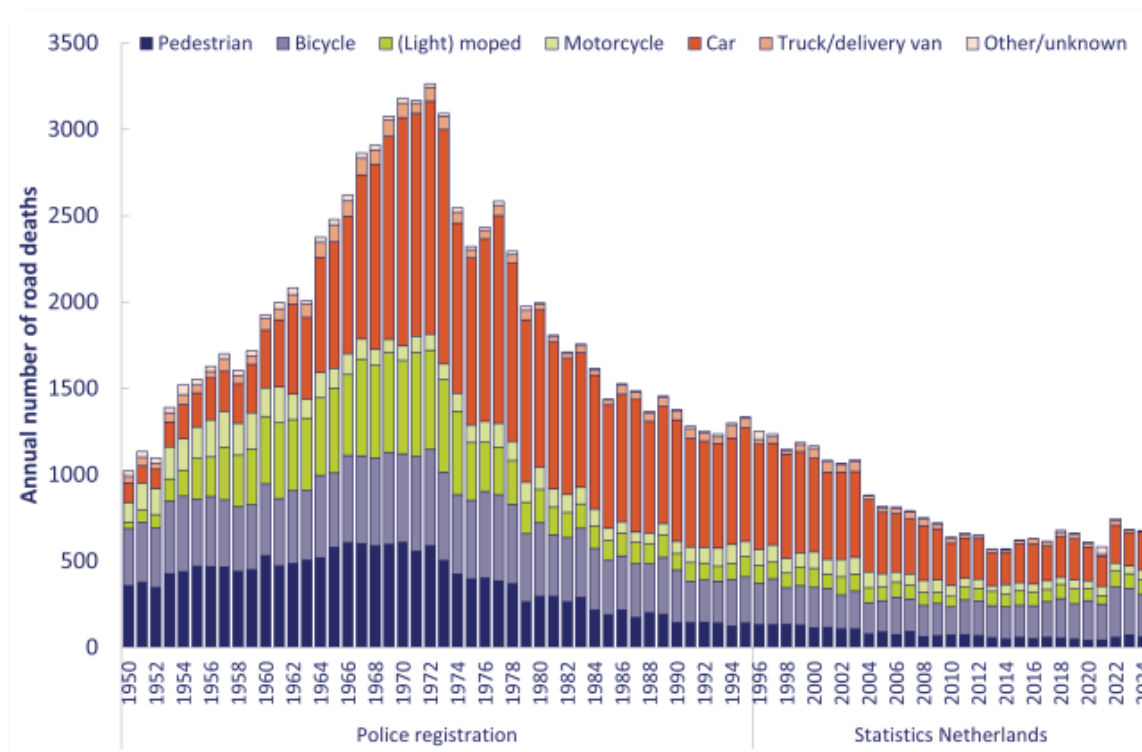


Figure 2. Road deaths in the Netherlands since 1950, by mode of transport. (Institute for Road Safety Research, 2024)

Chronic Damage

Car crashes remain a leading cause of death in many countries, but the health crisis tied to automobiles goes well beyond collisions. Chronic illnesses such as heart disease, respiratory issues, and certain cancers are linked to car-related pollution and sedentary lifestyles shaped by car dependency. Many deaths associated with transportation are not instantaneous but unfold over years due to inactivity and exposure to vehicle emissions (Litman, 2016). Figure 3 shows the top ten leading causes of potential human years of life

lost, with motor vehicle-related causes standing out. Litman (2016, p.5) notes that while these years lost are not solely due to car-centric policy but, are certainly exacerbated by it. Fishman and colleagues suggest that cycling has saved the Dutch about 6500 deaths annually and increased their lifespan by half a year (2015, p.14). These health benefits sit in stark contrast to the immediate health detriments of the health crisis in the Netherlands in the automobile era of the 1960s.

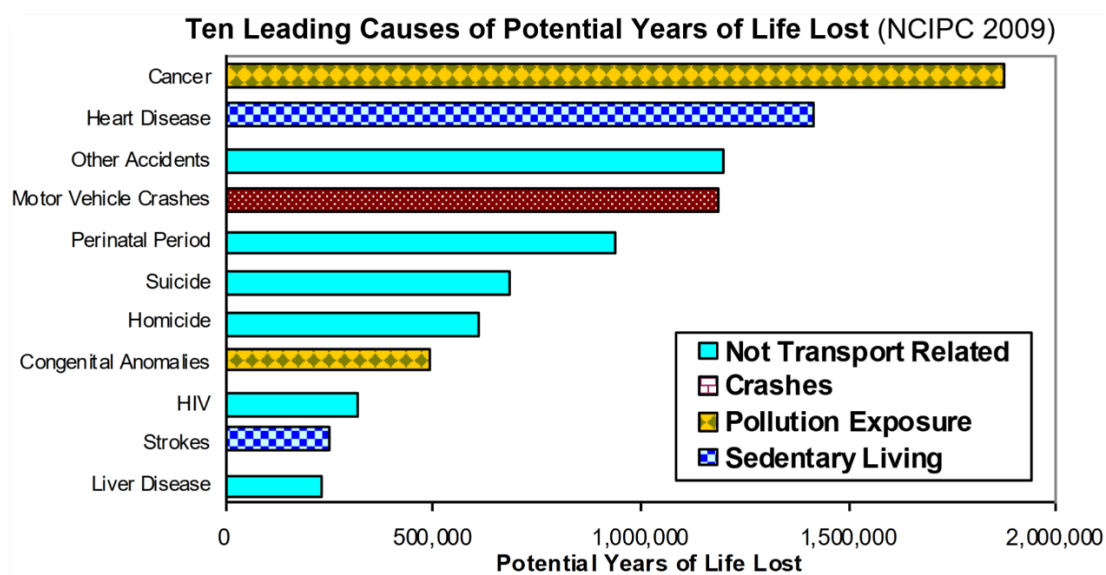


Figure 3. Health Implications from driving are not always instantaneous. (Litman, 2016).

Mobilizing Change

Grassroots Activism

Early detractors of the car voiced their concerns soon after the car came onto the road. The Netherlands saw a group who called themselves the Provos arise in the 1960s. Young radicals who were hungry for change, particularly a change away from overly capitalistic ideals of which the car symbolized. In an election, the Provos “won 13,000 votes

and one seat in the municipal elections in Amsterdam,” cementing their political power, however small (Furness, 2005, p. 411). Their loud anti-capitalistic rhetoric only led to a “mentality change rather than actual transformations of urban space” (Dekker, 2022, p. 222). Despite not making any policy changes, the Provos planted the seed of doubt among the Dutch who still believed in a car-centric Netherlands. It was not until the protests of the early 1970s that the government took notice in any meaningful way.

In the early 1970s, a Dutch Journalist, Vic Langenhoff’s 6-year-old daughter was killed in a traffic accident, and he was injured himself due to car-dominant planning (Dekker, 2022). He wrote an article decrying the public health crisis the Netherlands had found itself in the middle of. The headline of his piece, *Stop de Kindermoord* (Stop the Child Murder), captured the raw emotion of a grieving parent and a concerned citizen. Its bluntness struck a nerve, and activists quickly adopted the phrase, organizing protests on the roads in cities where children had been killed by cars (Bruno, Dekker, and Lemos 2021). The protesters were ordinary people like Maartje van Putten, a young mother, who had helped found the Stop the Child Murder group. This pressing topic had not been shunned by any political group and was a bipartisan issue everyone could get behind because, as Maartje van Putten declares, “there is not any politician, local, regional or central who can ignore people killed or wounded” (Bruno, Dekker, and Lemos 2021).

The tactics that Stop the Child Murder employed were quite effective. They used strong evocative language (as evidenced by their name) to make sure that these killings were not brushed away, not wrapped up in euphemism. They used children to bring to the protests and had the children make art about the dangers of crossing the road to evoke

strong emotions. They used the media to their advantage as much as they could to give this matter as many eyes as possible, combined with the aforementioned shock tactics. They were often confrontational with their local governments, making demands that they would try to hold their public servants to (Dekker, 2022). In 1972, they wrote to parliament demanding “police protection at intersections for school children ... more separate cycling and pedestrian facilities ... speed bumps at every intersection ... a ban on mopeds faster than 25km/h... complete a network of pedestrian and cycling facilities.” (Dekker, 2022). Despite these demands getting initially shot down, the message was starting to get across to the government that something had to be done.

Ultimately, these passionate demonstrations were not just spontaneous outbursts; they were a form of grassroots political organization that helped trigger real lasting change. Many people knew someone who had been injured or killed due to an automobile. It was a ubiquitous bipartisan social hatred that channeled strong protests into organized activism. The Stop the Child Murder movement saw citizens working with government officials and road engineers to achieve as few traffic-related deaths as possible. This movement, and others of the same goal, eventually swayed the government to progress policy towards a more cyclable and walkable country.

Dooievaar & Co-design

Public outrage alone did not change the Dutch health crisis. What made the Dutch case unique was that grassroots protest was met with mostly open ears in both the political and technical arenas. The movement to stop child traffic deaths in the Netherlands was

unique due to how receptive the government was to the will of the public. Many governmental leaders in the Netherlands at the time were amicable to environmentalism and were looking to make positive changes. It was this strong “transnational wave of environmental activism” which came about in part due to works like Jane Jacob’s 1961 book *The Death and Life of Great American Cities* and the 1972 report *The Limits to Growth* by non-profit organization The Club of Rome which criticized the current form of capitalism for the dangers it poses on communities (Bruno, Dekker, and Lemos 2021). To the activist groups like Stop the Child Murder, “the car symbolized everything that was wrong with capitalism,” while the bike was the symbol of a more equitable, environmentally friendly, economic system (Dekker, 2022, p. 224).

The activists’ message also reached students in technical fields. One such group from the Rotterdam Architecture Academy played a strategic role in releasing the Netherlands from its car dominance. Dooievaar, a group consisting of four architecture students studying at the Rotterdam Architecture Academy, was formed in 1970 in the city of The Hague (Verlaan, 2021). Like the Stop the Child Murder activist group, the name Dooievaar was supposed to shock the public. Dood, translating to dead, and ooievaar, translating to stork (The Hague’s symbol), was a warning to the government that the Hague would suffocate and die under the current dangerous car-centric planning (Dekker, 2022). These students had their minds on the same issues that Stop the Child Murder was working to solve, except they had a different angle. With their technical background, Dooievaar argued that the car-centric traffic schemes were “disintegrating the urban fabric” and likened them to a “spreading tumor that [will lead] to crude amputations and scars” (Verlaan, 2021, P. 149). Dooievaar was stalwart in its idea of involving the citizens at the policy planning level.

At the time, this idea was innovative, that citizens would have the right to “comment on already developed plans” and be included before the plans are set in motion (Bruno, Dekker, and Lemos 2021, p. 526). Dooievaar started with protests, building public support through media attention, and then were able to move on to directly contacting the government about their ideas (Dekker, 2022). These four students aided the pro-cyclist movement even further when they published a manual on how the average person could design their own cycling network. This manual contained a step-by-step process from sketching out a plan on paper to instructing the user on how to have their plan “enter into dialogue with local policymakers” (Dekker, 2022, p. 237). Dooievaar played a large role in the government taking an interest in the public and helped democratize urban planning in the Netherlands. Activist groups like Stop the Child Murder and Dooievaar provided the public with a channel and effective tactics through which they could direct their frustrations with the status quo in the 1970s Netherlands. Verlaan notes that a national survey in 1971 polled 64 percent of citizens in favour of “banning cars from city centers altogether” (2021, p. 143). This idea of banning cars from the city centers would later become a key Dutch traffic calming tool.

This shift in public sentiment coincided with a change in political leadership. Many of the initial changes made to policy were due to “the formation of a progressive ruling coalition in Dutch parliament” (Bruno, Dekker, and Lemos 2021). The Minister of Transport in 1971-72, one W. Drees Jr., understood that “different policy [was] necessary in order to “reduce the assault by road traffic on space and social climate” (Dutch Ministry of Transport, Public Works and Water Management 1999, p. 41). Everyone, the activists, the

engineers, and now the government, knew that something needed to be done, and there was a powerful catalyst right around the corner

Economic Catalyst: Oil Crisis

So far, only a few concrete moves had been made to pave the way for sustainable transportation institutionally. One large economic event was all the Netherlands needed to push them to seriously explore how car-free Dutch cities would feel. The 1973 oil crisis was a wake-up call for car-dependent countries like the Netherlands whose “transportation sector was particularly dependent on oil” (Hölsgens, 2019, p. 50). OAPEC sought to cut any country supporting Israel off from its oil supply. When oil prices suddenly spiked due to an embargo from OAPEC (Organization of Arab Petroleum Exporting Countries), it was not just a foreign policy issue. It became a daily, tangible disruption to people’s lives. For a country that relied heavily on imported oil (like many others), the economic impact was immediate and widespread. Driving quickly became unaffordable, and fuel had to be rationed. The government, in a scramble to cushion the blow on its constituents, came up with a car-free Sunday in which people would get around by other means. Church attendance “scarcely suffered” and “In front of every church stood masses of bicycles and mopeds” (Hellema et al., 2004, p. 107). The Dutch took this economic crisis in stride thanks to the preexisting robust cycling culture, which allowed for these Sunday church goers to use their bikes. The crisis also helped legitimize grassroots demands for more sustainable infrastructure, as concerns about energy security now aligned with existing calls for safer streets and cleaner cities. As policymakers began to rethink their own transportation priorities, the oil crisis

strengthened the argument that relying solely on fossil-fuel-driven mobility was both environmentally and economically unsustainable. In the years that followed, the experience of these Sundays helped generate broader political will to invest in cycling infrastructure and rethink national transport policy beyond the temporary emergency response.

Reclaiming the Country from the Car

Now that the public, government, and engineering experts had shared goals, they went to work devising a plan to make Dutch streets safer for everyone. The core principle of this new urban design is to restrict cars. During the 1970s, urban planners across the Netherlands were working hard to solve the problem of increased road fatalities. Innovative ways of restricting the automobile were popping up left and right. A new urban design tool was becoming more prevalent in the world of traffic engineering called traffic calming, which sought to slow down cars. A Dutch city named Delft was quick to act, designing in 1970 what would be the first speed bump in Europe (Schlabach 1997). This form of car-hostile infrastructure impeded cars' speed and forced drivers to pay closer attention to the road, with the goal of fewer pedestrian deaths.

Woonerf

The woonerf (translates to living yard) was one of the first Dutch traffic calming innovations. The woonerf came about when Dutch residents grew impatient with the government's delayed response to concerns of safety and pollution and took to the streets

to “tear up portions of the pavement on their street,” forcing cars to “maneuver around the resulting obstruction at low speed” (Lydon and Garcia 2015, p. 28). The idea is to design a local road in such a way that the design itself lets the driver know that they are guests in the woonerf. Although signs are used, the qualities of the woonerf are enough to slow the driver down. The woonerfs being narrow and obstacle-laden signal to the driver that they should slow down, drive attentively, so as not to damage their vehicle or another’s property. The result of this design is a space where “pedestrians and children have greater rights to residential street space than traffic” (Appleyard, 1980, p. 113). The woonerf reflects Schiller and Kenworthy’s core characteristics of “accessibility and quality,” since the woonerf ensures that everyone, not just those in cars, can safely access and enjoy the space, while also improving the quality of life through careful design that promotes the health and safety of the community. The initial guerrilla woonerf in Delft was not seen as a problem by the municipal government, and years later, in 1976, the Dutch Parliament officially adopted the idea into street design standards due to how successful it was (Lydon and Garcia 2015, p. 28). The woonerf spread across the Netherlands and helped make streets safer.

Policy

It was right around 1974-1976 that the first wave of car-restricting policy was being implemented in the Netherlands. Since fewer people were using cars, parking lots were able to be reclaimed and “municipalities ceased to turn their historic squares into parking lots” (Bruno, Dekker, and Lemos 2021, p. 527). Streets were being given back to the people and away from the auto. Dutch policy was written in such a way that lowering the speed

limit from 50 km/h to 30 km/h for residential neighborhoods was effortless because it did not have to go through parliament, and was something that the transport minister was able to do. With the country's decentralized approach to road governance, local municipalities had significant freedom to design neighborhood streets without strict national restrictions (Bruno, Dekker, and Lemos, 2021).

Certain cities even started deconstructing many of the through-roads for cars, forcing them to take the longer and less efficient route. This further discouraged the use of the car and encouraged cycling. The key with many of these changes was that they were systemic. They changed the systems that facilitated the rise in crashes rather than encouraged individual change. One could chase “crash-prone” drivers all day long, but as Tiwari and Mohan put it, “[humans are] physically vulnerable and, moreover, make errors and commit offences” (2021). When we fool-proof everyday mobility systems, that is when we see a decline in traffic-related fatalities. Looking back at Figure 2 we can see a slight downturn in traffic deaths in 1973 and then a sharp decline in traffic deaths in 1974, indicative of fewer people using cars and traffic safety policy working. Throughout the 1970s, accidents were never higher than they were in 1973 and generally trended downwards despite there being more cars on the road. All these efforts proved to be worthwhile. Even the “policymakers in the mid-1970s had abandoned the exclusive focus on building highways in response to predictions of ever-growing car traffic,” instead coming up with a long-term cycling infrastructure plan (Ploeger and Oldenziel, 2020, p. 15). See Appendix A for more examples of Dutch car-restricting urban design that contributed to this shift.

What Can Other Countries Learn?

The Dutch transition away from car-centric planning offers important lessons for the future of urban mobility worldwide. As cities continue to face challenges such as traffic congestion, air pollution, and rising greenhouse gas emissions, the Netherlands provides a clear example of how a shift toward sustainable transportation can be both effective and widely accepted. Countries like the United States are in dire need of a transportation reform like the Dutch had in the 70s. Figure 4 shows the total number of road deaths trending upwards to numbers similar to 20 years ago, showing no progress being made. Once the Dutch got the ball rolling on restricting cars, they consistently lowered the total number of road deaths despite their population increasing each year (see Figure 2). The upward trend in the USA suggests that despite advances in vehicle technology and enforcement efforts, the country continues to prioritize car-oriented infrastructure at the expense of safety. The number of fatalities is higher as well. Pucher and Dijkstra (2003) found that when comparing the two countries the “American bicyclists are ... over 3 times as likely [to get

killed] as Dutch cyclists” (p.1511). One lesson is how the onus of traffic safety should be passed from the individual to the institution

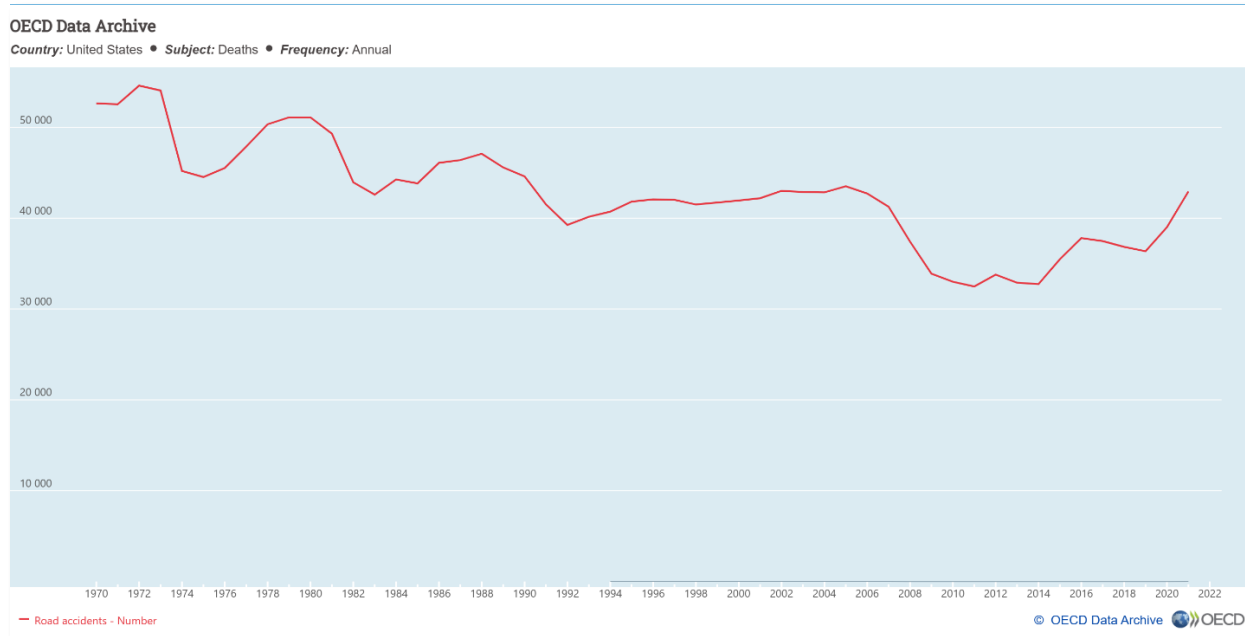


Figure 2. United States Road Deaths have been trending upwards in the last 10 years (OECD Data Archive).

The Netherlands’ transformation was not the result of a single policy, but rather the outcome of coordination between grassroots activism, political support, and innovative urban planning. Future transportation strategies should recognize that automobiles are only one of many mobility options. Investments in cycling, walking, and public transit can create safer, more inclusive cities while improving public health and reducing environmental harm. Dutch tools such as traffic-calmed streets and neighborhood-oriented design can serve as practical models for cities looking to reclaim urban space from vehicles and return it to people. However, these changes require consistent political commitment, active public engagement, and a willingness to challenge car-dominant narratives. The

Dutch experience shows that it is possible to reimagine the structure of cities by placing human needs ahead of traffic flow.

The Netherlands' experience demonstrates how lasting change requires systemic shifts in both infrastructure and culture. One clear example of this long-term success is the widespread adoption of multimodal transport. Today, public transport and cycling are not competing modes; rather, they are complementary. As Van Der Craats et al. (2025) note, "Individuals with public transport season tickets were significantly more likely to use shared bikes, likely as a first/last mile solution to public transport stations" (p. 4). This kind of seamless, multimodal integration reflects how deeply sustainable mobility is now embedded in Dutch urban life. It shows that when walking, cycling, and transit are properly connected, people naturally adopt lower-emission, people-first forms of transportation.

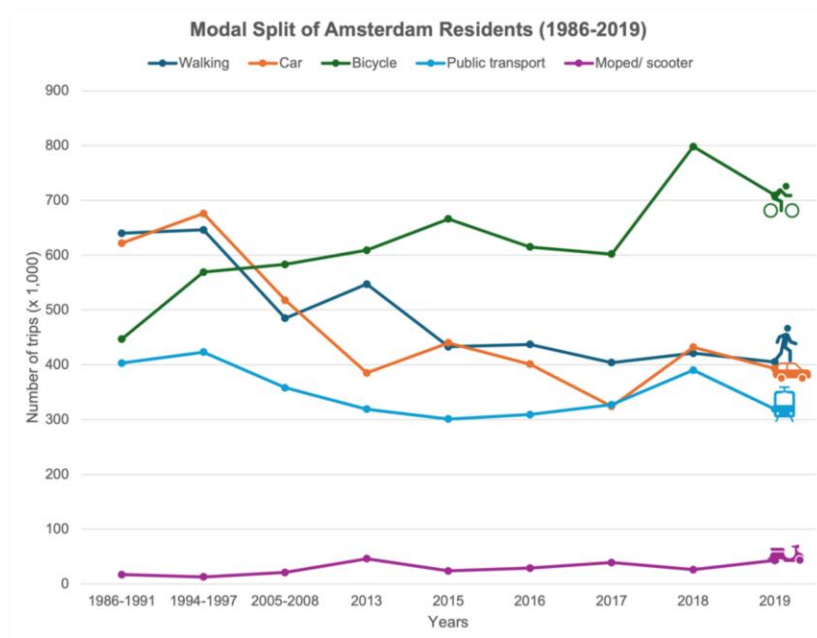


Figure 3. Cycling has the most total number of trips taken (Modified by van der Craats & van Lierop, n.d., from: Gemeente Amsterdam, 2021; Ministerie van Infrastructuur en Waterstaat, 2017).

Conclusion

The Dutch response to the health crisis of the 1970s stands as one of the most compelling examples of how cities can reclaim their streets through a combination of grassroots action, institutional reform, and visionary design. This transformation was not driven by isolated events, but by a convergence of public outrage, political will, technical innovation, and economic necessity. What began as grief over child deaths catalyzed a national reckoning with car-centric urban planning and led to the creation of safer, more equitable, and more sustainable cities. This paper has shown how the bicycle, often overlooked in policy discussions, became both a practical solution and a powerful symbol of the Dutch transition. Through the triple bottom line framework, cycling emerged not only as a low-emission mode of transport but as a force for community cohesion and economic efficiency. The widespread adoption of cycling was not accidental; it was cultivated through early advocacy, national consensus-building, and policy decisions that reshaped the urban environment to support active travel.

Critically, the Dutch did not rely on individual behavior change alone. They redesigned systems: slowing traffic, narrowing streets, and giving pedestrians and cyclists legal and spatial priority. From citizen-led protests like *Stop de Kindermoord* to design innovations like the *woonerf*, these reforms fundamentally shifted the expectations placed on public space. As Tiwari and Mohan point out, humans are inherently fallible, so it is urban systems and not individual choices that must be made safer. What solidified these efforts was a culture of cooperation. The unique Dutch *polder* model fostered collaborative decision-making between citizens, planners, and politicians. Student groups like *Dooievaar* illustrated that design could be a tool of democratic resistance, empowering people to not

only critique but co-create the urban landscape. When the 1973 oil crisis struck, this infrastructure and civic mindset were already in place, allowing the Netherlands to adapt swiftly, reinforcing the case for long-term sustainable mobility.

The legacy of this transformation endures. Today, Dutch cities feature deeply integrated multimodal systems, where cycling and transit complement one another (see Figure 5). As Van Der Craats et al. (2025) observed, the use of shared bikes as first/last-mile transit solutions illustrates how cultural change has become embedded in everyday routines. This is the outcome of decades of sustained, systemic reform. For other countries facing similar challenges, like rising road deaths, climate pressures, and social fragmentation, the Dutch case offers more than inspiration. It offers proof that car dominance is not a fixed reality, but a policy choice. If we center people over throughput, cooperation over convenience, and long-term wellbeing over short-term gain, then safer, healthier, and more just streets are not only possible... they are inevitable.

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Appendix A: Netherlands modal equity and cyclist priority in Leeuwarden.



Figure 4. Separated bike path



Figure 5. School children cycling together with supervisors.



Figure 6. Multi-use pathway with plenty of bike storage on the left.



Figure 7. Car restrictive road featuring a single car lane with two bike lanes.